



Measuring Domestic Output and National Income Assessing the Economy's Performance

National income accounting measures the economy's overall performance. It does for the economy as a whole what private accounting does for the individual firm or for the individual household. A business firm measures its flows of income and expenditures regularly—usually every 3 months or once a year. With that information in hand, the firm can gauge its economic health. If things are going well and profits are good, the accounting data can be used to explain that success. If things are going badly and profits are poor, the firm may be able to identify the reason by studying the record over several accounting periods. All this information helps the firm's managers plot their future strategy.

This accounting enables economists and policymakers to:

1. Assess the health of the economy by comparing levels of production at regular intervals.
2. Track the long-run course of the economy to see whether it has grown, been constant, or declined.
3. Formulate policies that will safeguard and improve the economy's health.

Gross Domestic Product The primary measure of the economy's performance is its annual total output of goods and services or, as it is called, its *aggregate output*. There are several ways to measure aggregate output depending upon how one wishes to define "an economy."

Gross domestic product (GDP) defines aggregate output as the dollar value of all final goods and services produced within the borders of a country during a specific period of time, typically a year. Under this definition, the value of the cars produced at the Toyota factory in Ohio clearly count as part of U.S. aggregate output rather than Japanese aggregate output because the cars are made within the borders of the United States.

Avoiding Multiple Counting

To measure aggregate output accurately, all goods and services produced in a particular year must be counted once and only once. To avoid counting those components each time, GDP includes only the market value of *final goods* and ignores *intermediate goods* altogether.

Intermediate goods are products that are purchased for resale or further processing or manufacturing. Crude oil, Steel beams are intermediate goods, Lettuce, carrots, and vinegar in restaurant salads are intermediate goods,

Final goods are products that are purchased by their end users, gasoline used for personal transportation is a final good and completed high-rise apartments are final goods.

Alternatively, we could avoid multiple counting by measuring and cumulating only the *value added* at each stage. **Value added** is the market value of a firm's output *less* the value of the inputs the firm has bought from others.

CONCEPTS OF NATIONAL INCOME

The important concepts used about national income are:

1. GDP (Gross Domestic Product)
2. GNP (Gross National Product)

3. NNP (Net National Product) 4. NI (National Income)

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5. PI (Personal Income)

6. DI (Disposable Income)

Gross Domestic Product (GDP)

It is the first measure for national income. Gross Domestic Product (GDP) of a country means total value of goods and services produced within the country during one year. The concept of GDP ignores the incomes received by its nationals from their property abroad or their services in other countries. Usually two kinds of firms work in a country. The local firms have their origin (or head office) in the same country. Foreign firms, which belong to other countries, have their origin and head office somewhere abroad. For example SERVIS (shoes) is a Pakistani firm while BATA is a foreign firm working in Pakistan. Similarly the individuals working in a country will be either locals or foreigners. Total value of output by all kinds of firms, which they produce inside Pakistan, is included in GDP of Pakistan. Thus production of both SERVIS and BATA is included in GDP of Pakistan. The salary of a Japanese engineer working in HONDA factory Lahore is also included in our GDP, since he performs his services in Pakistan. On the other hand, income of a Pakistani working in DUBAI is not counted in our GDP since the services have been rendered outside Pakistan.

Gross National Product (GNP)

GNP is the most important and commonly used concept in national accounting. It is a basic but rough estimate of economic performance of a nation. GNP measures the total value of output of goods and services produced by the people in or outside their country. GNP is a wider and more comprehensive term than GDP. In addition to domestic production it considers net income received from abroad. **It may be defined as:**

“GNP of a country is the annual sum of market values of all final goods and services produced by its nationals in or outside their country.”

The meaning of GNP is broader than GDP, which refers to only total domestic production inside a country. GNP includes all production by the residents of a country whether it is done in the country or somewhere abroad. The GNP of a country, say Pakistan, includes the profits sent home by all Pakistani firms and incomes of individuals even if they are working outside Pakistan. Similarly the part of profits and incomes transferred abroad by foreign firms and individuals are excluded.

$GNP = GDP + \text{income received by Pakistanis from abroad} - \text{income of foreigners taken away from Pakistan.}$ OR

$GNP = GDP + \text{Net Factor Income from abroad.}$

Real GNP vs. Nominal GNP:

If GNP is measured in current prices, it is called nominal GNP. By using the index of the general price level, nominal GNP is deflated and converted into prices of some base year. The GNP

adjusted for price changes is called real GNP or GNP at constant prices. $\text{Real GNP} = \text{Nominal GNP} / \text{deflator}$.

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Net National Product (NNP) (National Income at market prices):

NNP is the measure of national production of a country obtained by deducting the amount of depreciation from GNP. NNP is a better measure to know how much total production an economy can make available for use without decreasing its capital stock. **$NNP = GNP - \text{Depreciation}$**
NNP is the true measure of national income of a country.

National Income (at Factor Cost):

National income is the total of all incomes earned by the factors of production in the form of wages, rent, interest and profit.

NNP gives us a measure of national income by adding the money values of all goods and services produced by a country in a given period at market prices. **$\text{National income} = NNP - \text{indirect taxes} + \text{subsidies}$**

Personal Income (PI):

Personal income is the total income received by the people from all sources. Personal income differs from national income for two reasons.

Firstly, there are some deductions from the earnings of persons in the form of social security contributions before they receive their salaries. Then the limited companies do not distribute the whole of the earned profit among the shareholders. They keep a part of it as reserves. Some part of the profits of firms is also taken away by the government as corporate taxes.

Secondly, there are some payments received by the people which they have not earned as incomes in exchange for production of goods e.g. pensions, gifts, Zakat, alms, unemployment allowance, scholarship, etc. These payments are called transfer payments and have to be added to national income to arrive at personal income.

$\text{Personal income (PI)} = \text{NI} - \text{Social security contributions}$

- Corporate taxes
- Undistributed profits (reserve fund) + Transfer payments.

Disposable Income (DI) (or Disposable Personal Income — DPI):

Disposable income (or disposable personal income) is the total net amount left with the individuals and households when they have paid direct taxes **Example:**

$\text{Disposable income (DI)} = \text{PI} - \text{Personal taxes}$

PER CAPITA INCOME (Average income):

In order to know the average income of people in a country, per capita income is calculated i.e. income per head of population. It is obtained when national income is divided by the population of the country.

Per capita income = national income/ population

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The concept of per capita income is used to have an idea of the average standard of living of the people of a country.

A higher per capita income shows that people have more to consume in terms of goods and services. Lower per capita income implies widespread poverty.

Nominal GDP vs. Real GDP:

Nominal GDP is the total value of all goods and services produced in a given time period, usually quarterly or annually. **Real GDP is nominal GDP adjusted for inflation.** Real GDP is used to measure the actual growth of production without any distorting effects from inflation. Real GDP tracks the total value of goods and services calculating the quantities but using constant prices that are adjusted for inflation. This is opposed to nominal GDP that does not account for inflation.

Further definitions & concepts:**Public saving**

Public saving can be defined as a difference between government revenue and government expenditure.

Corporate saving

“Corporate Saving” is the difference between income and consumption.

Transfer payments

A payment made or income received in which no goods or services are being paid for, such as a benefit payment or subsidy is called transfer payment

Depreciation

Gradual decrease in the value of an asset is called depreciation.

Value added

The amount by which the value of an article is increased at each stage of its production, exclusive of initial costs is called value added.

Subsidy

A sum of money granted by the state or a public body to help an industry or business keep the price of a commodity or service low.

Consumption (C):

It is regarded as total expenditure by households on goods and services which yield utility in the current period.

Savings (S) are that part of the disposable income which is not spent in the current period. It follows that disposable income ($Y - T$) minus saving equals consumption.

Investment (I) is the production of or expenditure by firms on goods and services which are not for current consumption: that is, real capital goods, like factors-machines, bridges and motorways, all goods which yield a flow of consumer goods and services in future period.

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Two Ways of Looking at GDP:

a. Spending and Income

One way is to see how much the final user paid for it. That will tell us the market value of the final product. Or we can add up the entire wage, rental, interest, and profit incomes that were created in producing the coat. The second approach is the value-added technique used in Table 27.2.

TABLE 27.2 Value Added in a Five-Stage Production Process

(1) Stage of Production	(2) Sales Value of Materials or Product	(3) Value Added
	\$ 0	
Firm A, sheep ranch	120	\$120 (= \$120 - \$ 0)
Firm B, wool processor	180	60 (= 180 - 120)
Firm C, coat manufacturer	220	40 (= 220 - 180)
Firm D, clothing wholesaler	270	50 (= 270 - 220)
Firm E, retail clothier	350	80 (= 350 - 270)
Total sales values	\$1,140	
Value added (total income)		\$350

The final-product approach and the value-added approach are two ways of looking at the same thing. What is spent on making a product is income to those who helped make it. If \$350 is spent on manufacturing a coat, then \$350 is the total income derived from its production.

How to measure GDP:

We can look at GDP in the same two ways. We can view GDP as the sum of all the money spent in buying it. That is the *output approach*, or **expenditures approach**. Or we can view GDP in terms of the income derived or created from producing it. That is the *earnings* or *allocations approach*, or the **income approach**.

As illustrated in Figure 27.1, we can determine GDP for a particular year either by adding up all that was spent to buy total output or by adding up all the money that was derived as income from its production. Buying (spending money) and selling (receiving income) are two aspects of the same transaction. On the expenditures side of GDP, all final goods produced by the economy are bought either by three domestic sectors (households, businesses, and government) or by foreign buyers. On the income side (once certain statistical adjustments are made), the total receipts acquired from the sale of that total output are allocated to the suppliers of resources as wage, rent, interest, and profit.

The Expenditures Approach

To determine GDP using the expenditures approach, we add up all the spending on final goods and services that has taken place throughout the year. National-income accountants use precise terms for the types of spending listed on the left side of Figure 27.1.





Personal Consumption Expenditures (C)

What we have called “consumption expenditures by households,” the national income accountants call **personal consumption expenditures**. This term covers all expenditures by households on goods and services.

Gross Private Domestic Investment (Ig)

Under the heading **gross private domestic investment**, the accountants include the following items:

- All final purchases of machinery, equipment, and tools by business enterprises.
- All construction.
- Changes in inventories.
- Money spent on research and development (R&D) or for the creation of new works of art, music, writing, film, and so on.

Government Purchases (G)

The third category of expenditures in the national income accounts is **government purchases**, officially labeled “government consumption expenditures and gross investment.” These expenditures have three components: (1) expenditures for goods and services that government consumes in providing public services; (2) expenditures for *publicly owned capital* such as schools and highways, which have long lifetimes; and (3) government expenditures on R&D and other activities that increase the economy’s stock of know-how.

Net Exports (Xn)

International trade transactions are a significant item in national income accounting.

Accountants simplify this formula for GDP by defining **net exports**, X_n , to be equal to exports minus imports:

Net exports (X_n) = exports (X) – imports (M)

Using this definition of net exports, the formula for gross domestic product simplifies to, $GDP = C + I_g + G + X_n$

The Income Approach

The income approach measures GDP by summing the incomes that firms pay households for the services of the factors of production they hire—wages for labor, interest for capital, rent for land, and profit for entrepreneurship.

The following are the reward, which are added up:

- **Wages:**
All wages salaries and other monetary rewards, which employs receive in a year, including compensation, paid to them.
- **Interest:**
All types of interest, received by the investors during the year on bond and loans
- **Rent:**
Rent of all lands buildings and other properties, which the owner receives in a year.
- **Profit:**
It includes the net corporate earnings of the properties including dividends paid to the shareholders, undistributed profit of the company's corporate taxes which are paid to the govt.



Product method (NI at market prices):

According to this method the economy is divided into different sector such as agriculture, manufacturing, transport and communication mining and services etc. The net value of all the production of these sectors is added in a given year. The income form aboard is also included this gives us gross national product. When the amount of depreciation is subtracted we get the net national product (NNP) or national income.

FIGURE 27.1 The expenditures and income approaches to GDP. There are two general approaches to measuring gross domestic product. We can determine GDP as the value of output by summing all expenditures on that output. Alternatively, with some modifications, we can determine GDP by adding up all the components of income arising from the production of that output.

Expenditures, or output, approach

Consumption expenditures by households
plus
Investment expenditures by businesses
plus
Government purchases of goods and services
plus
Expenditures by foreigners

= GDP =

Income, or allocations, approach

Wages
plus
Rents
plus
Interest
plus
Profits
plus
Statistical adjustments

CIRCULAR FLOW OF INCOME

Circular flow of income means that income and expenditure in an economy are related to each other in a circular way.

Definition:

Circular-flow diagram presents a visual model of the economy which highlights the relationship between income generating and income spending activities. We know that receiving income and production of goods are two sides of one activity. There will be income only if goods and services are produced.

Relation of Spending & Earning:

When we study the process of how income is received and how it is spent, we find that there is a direct link between production of goods or services and spending of incomes. To explain it we take a simple model of economy, which consists of two types of decision makers, firms and households.

Circular Flow of Income in a Two -Sector Economy:

According to circular flow of income in a two -sector economy, there are only two sectors of the economy, i.e., household sector and business sector. Government does not exist at all, therefore, there is no public expenditure, no taxes, no subsidies, no social security contribution, etc. The

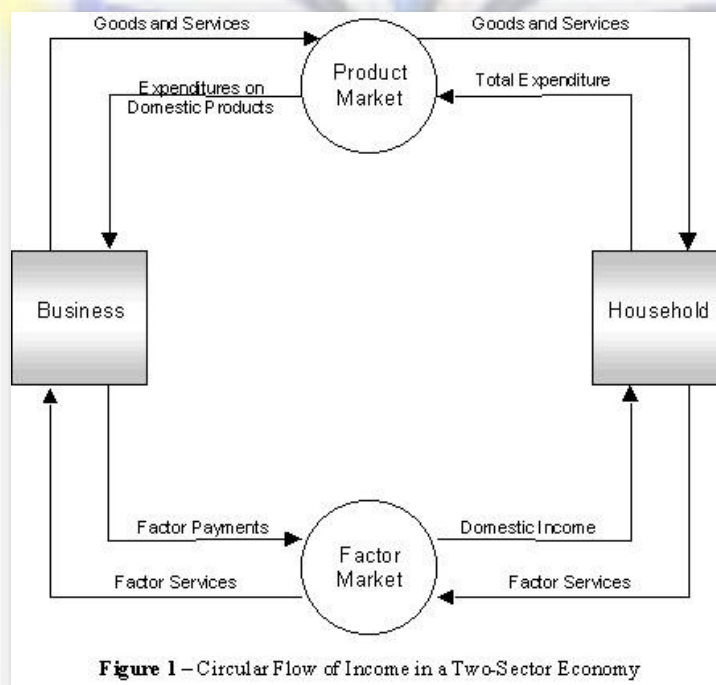
economy is a closed one, having no international trade relations. Now we will discuss each of the two sectors:

(i) **Household Sector:** The household sector is the sole buyer of goods and services, and the sole supplier of factors of production, i.e., land, labour, capital and organisation. It spends its entire income on the purchase of goods and services produced by the business sector. Since the household sector spends the whole income on the purchase of goods and services, therefore, there are no savings and investments. The household sector receives income from business sector by providing the factors of production owned by it.

(ii) **Business Sector:** The business sector is the sole producer and supplier of goods and services. The business sector generates its revenue by selling goods and services to the household sector. It hires the factors of production, i.e., land, labour, capital and organisation, owned by the household sector. The business sector sells the entire output to households. Therefore, there is no existence of inventories. In a two-sector economy, production and sales are thus equal. So long as the household sector continues spending the entire income in purchasing the goods and services from the business sector, there will be a circular flow of income and production.

Flow of services and Money:

- (i) Real flow Services of labour, land, and capital going from households to firms, and products of firms as physical goods or services flowing to households.
- (ii) Money flow Firms making payments to households for factor services and households spending money to buy goods from firms.



GDP Price Index

In the actual economy, a **price index** is a measure of the price of a specified collection of goods and services, called a “market basket,” in a given year as compared to the price of an identical (or highly similar) collection of goods and services in a reference year. That point of reference, or benchmark, is known as the base period, **base year**, or, simply, the reference year. More formally,

$$\text{Price index in given year} = \frac{\text{price of market basket in specific year}}{\text{price of same market basket in base year}} \times 100$$

$$\text{Price index, year 2} = \frac{\$20}{\$10} \times 100 = 200$$

$$\text{Price index, year 3} = \frac{\$25}{\$10} \times 100 = 250$$

TABLE 27.6 Calculating Real GDP (Base Year = Year 1)

Year	(1) Units of Output	(2) Price of Pizza per Unit	(3) Price Index (Year 1 = 100)	(4) Unadjusted, or Nominal, GDP, (1) × (2)	(5) Adjusted, or Real, GDP
1	5	\$10	100	\$ 50	\$50
2	7	20	200	140	70
3	8	25	250	200	80
4	10	30	—	—	—
5	11	28	—	—	—

How to calculate real GDP:

$$\text{Real GDP} = \frac{\text{nominal GDP}}{\text{price index (in hundredths)}} \quad (2)$$

Income & Growth:**Economic Growth**

- An increase in real GDP occurring over some time period.
- An increase in real GDP per capita occurring over some time period.

With either definition, economic growth is calculated as a percentage rate of growth per quarter (3-month period) or per year. For the first definition, for example, real GDP in the United States was \$15,961.7 billion in 2014 and \$16,348.9 in 2015. So the U.S. economic growth rate for 2015 was 2.4 percent



The second definition of economic growth in the bulleted list takes into consideration the size of the population. **Real GDP per capita** (or per capita output) is the amount of real output per person in a country. It is calculated, as follows.

$$\text{Real GDP per capita} = \frac{\text{Real GDP}}{\text{Population}}$$

Arithmetic of Growth

The mathematical approximation called the **rule of 70** provides a quantitative grasp of the effect of economic growth. The rule of 70 tells us that we can find the number of years it will take for some measure to double, given its annual percentage increase, by dividing that percentage increase into the number 70:

$$\text{Approximate number of years required to double real GDP} = \frac{70}{\text{annual percentage rate of growth}}$$

Determinants of Growth

These determinants of economic growth can be grouped into four supply factors, one demand factor, and one efficiency factor. **Supply Factors**

The first four determinants of economic growth relate to the physical ability of the economy to expand. They are:

- Increases in the quantity and quality of natural resources.
- Increases in the quantity and quality of human resources.
- Increases in the supply (or stock) of capital goods.
- Improvements in technology. **Demand Factor**

The fifth determinant of economic growth is the **demand factor**:

- To actually achieve the higher production potential created when the supply factors increase or improve, households, businesses, and the government must also expand their purchases of goods and services so as to provide a market for all the new output that can potentially be produced.

Efficiency Factor

The sixth determinant of economic growth is the **efficiency factor**:

- To reach its full production potential, an economy must achieve economic efficiency as well as full employment.

Aggregate Demand & supply:

Aggregate supply and demand refers to the concept of supply and demand but applied at a macroeconomic scale. Aggregate supply and aggregate demand are both plotted against the aggregate price level in a nation and the aggregate quantity of goods and services exchanged at a specified price.

Aggregate Supply

The aggregate supply curve measures the relationship between the price level of goods supplied to the economy and the quantity of the goods supplied. In the short run, the supply curve is fairly elastic, whereas, in the long run, it is fairly inelastic (steep). This has to do with the factors of production that a firm is able to change during these two different time intervals.

Aggregate Demand

Since consumer demand does not face the same constraints faced by suppliers, there is no relative change in the elasticity of demand itself. Rather, the steepness of the demand curve depends on the price elasticity of demand for the good. Thus, the aggregate demand curve follows a consistent downward slope, whose elasticity is subject to change due to factors such as:

- Changing consumer preferences
- New literature about certain products
- Changes in the rate of inflation
- Changes in interest rates
- Changes in the level of household wealth
- Foreign currency risk

Aggregate Demand - Aggregate Supply Model

The AD-AS model is a graphical representation of the aggregate demand and aggregate supply within an economy. It was founded by famous economist John Maynard Keynes. Economists will often use this model to illustrate how aggregate supply and demand change as the economy booms and recedes.

The model is also used to illustrate if the economy is producing in a:

- surplus
- deficit
- equilibrium

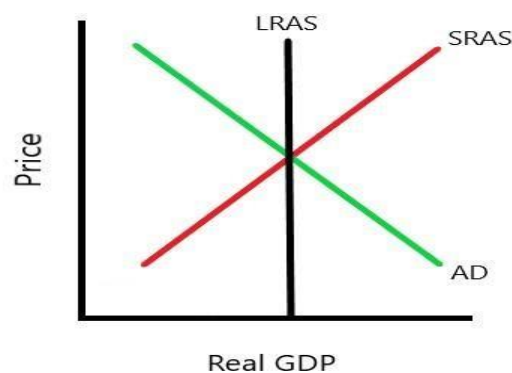
There are a couple of critical concepts that one must understand to comprehend the breadth of the **aggregate demand - aggregate supply model**. The long-run aggregate supply (LRAS) curve of the model is a vertical line. This is due to how, in Keynesian economics, economies are elastic in the long run. This means that as an economy determines what goods are in demand, it can perfectly adjust if given enough time. As the long-run aggregate supply curve, it is thus illustrated as a vertical line that goes through the supply and demand curves. Since John Maynard Keynes believed that economies are self-sufficient in the long run, the vertical LRAS further depicts how perfectly elastic aggregate supply is.

Features of Aggregate Demand-Aggregate Supply Model

The aggregate demand and aggregate supply model has certain key features that must be taken into consideration to further understand the forces of supply and demand. If there is too much demand

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in an economy, the short-run equilibrium will be ahead of the long-run aggregate supply. As a result, gaps form to the right of the long-run aggregate supply curve. These are referred to as inflationary gaps. The AD-AS model demonstrates the relationship between long-run economics and short-run economics.



The intersection of LRAS, SRAS, and AD is the market equilibrium.

Public Finance: Expenditures and Taxes Government and the Circular Flow

In Figure 20.1, we integrate government into the circular flow model first shown in Figure 2.2. Here flows (1) through (4) are the same as the corresponding flows in that figure.

Flows (1) and (2) show business expenditures for the resources provided by households. These expenditures are costs to businesses but represent wage, rent, interest, and profit income to households. Flows (3) and (4) show household expenditures for the goods and services produced by businesses.

Now consider what happens when we add government. Flows (5) through (8) illustrate that government makes purchases in both product and resource markets. Flows (5) and (6) represent government purchases of such products as paper, computers, and military hardware from private businesses.

Flows (7) and (8) represent government purchases of resources. The federal government employs and pays salaries to members of Congress, the armed forces, Justice Department lawyers, meat inspectors, and so on. State and local governments hire and pay teachers, bus drivers, police, and firefighters. The federal government might also lease or purchase land to expand a military base and a city might buy land on which to build a new elementary school.

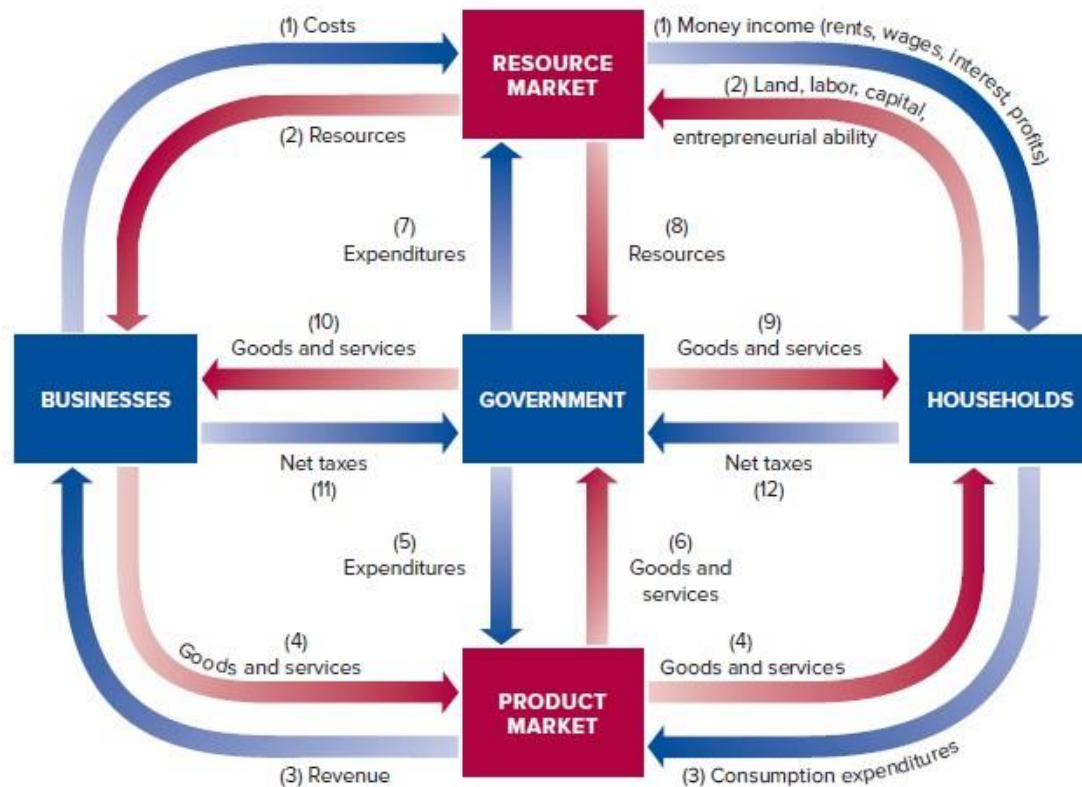
Government then provides goods and services to both households and businesses, as shown by flows (9) and (10). Governments rely on three revenue sources to finance those goods and services: taxes, borrowing, and the proprietary income generated by government-run or governmentsponsored businesses like public utilities and state lotteries. These revenues flowing

from households and businesses to government are included in flows (11) and (12), which are labeled as “net taxes” for two reasons. First, the vast majority of the money raised by these three revenue sources comes from taxes; thus, it is sensible to have these labels refer to taxes. Second, the labels refer to *net* taxes to indicate that they also include “taxes in reverse” in the form of transfer payments to households and subsidies to businesses. Thus, flow (11) entails various subsidies to



farmers, shipbuilders, and airlines as well as income, sales, and excise taxes paid by businesses to government.

FIGURE 20.1 Government within the circular flow diagram. Government buys products from the product market and employs resources from the resource market to provide goods and services to households and businesses. Government finances its expenditures through the net taxes (taxes minus transfer payments) it receives from households and businesses.



Government Finance

Public Finance:

Prof. Dutton says, “Public finance deals with the income and expenditures of public authorities and their adjustments”.

2. Private Finance:

“Private finance deals with the income and expenditure of a person or individual”.

Sources of Public Revenue:

1. Tax:

Tax means “Burden and Duty”. A compulsory contribution made by the public for the support of a Govt. These are the kinds of Tax:

2. Income tax and wealth tax:

Income Tax is the form of direct tax in which taxpayer assessed his total income wealth. Tax imposed by tax authorities under self-assessment system.

3. Sales Tax:

This is form of indirect taxes and collected by the excise department. Tax which imposed on the value of goods supplied known as sales tax.

4. Custom:



Tax which imposed an import or export of goods known as customer duty or tax.

5. Excise duty:

The Federal excise duty imposed upon locally manufactured goods and services.

6. Fines:

If anyone break any law of state then Govt, imposed on him some amount as penalty. This one is the big source of income of Govt.

7. Fee:

The Govt, provides some services to the people and dard money in return for these services e.g. educational institutes fee, license fee etc.

8. Local rates:

Tax levied by local Govt, bodies known as local rates e.g. Municipal Corporation, town committees, district council and tehsil councils etc.

9. Progressive tax:

If the rate of tax increase as the taxable income increase or rate of tax decrease as the taxable income decrease, the tax is called progressive tax. So, the principle of progressive tax is,

- Higher the income, higher the rate of tax.
- Lower the income, lower the rate of tax

10. Regressive tax:

Regressive tax system is opposite of progressive tax system. In progressive taxes, rate of tax decreases as taxable income increase or rate of tax increases as taxable income decrease. So, the principle of regressive tax is:

- Higher the income, lower the rate of tax.
- Lower the income, higher the rate of tax.

11. Proportional tax:

Proportional taxes are also called flat rate taxes because these are imposed at uniform rate at all income levels. Proportional tax is a tax which is paid at the same rate whether person's income increase or decrease.

12. Digressive tax:

It is a tax system in which tax rate is proportional up to a certain limit of income and after that limit it become progressive. So, it is the combination of proportional and progressive taxes.

13. Value added tax:

A tax system in which govt, collects tax at each stage of production is called value added tax. **For Example:** Cotton passes through six stages before retails sells readymade garments of cotton. So, value added tax is received at every stage of tax addition:

14. Advaloran tax:

It is a tax which is imposed on a commodity according to its value is called advaloren tax. For example, 15% import duty on electronic goods.

“Tax incidence”

Incidence of tax is an economic term for understanding the division of a tax burden between stakeholders, such as buyers and sellers or producers and consumers. Tax incidence can also be

related to the price elasticity of supply and demand. When supply is more elastic than demand, the tax burden falls on the buyers. If demand is more elastic than supply, producers will bear the cost of the tax.





Efficiency Loss Of Taxation?

The term deadweight loss of taxation refers to the measurement of loss caused by the imposition of a new tax. This results from a new tax that is more than what is normally paid to the government's taxing authority. This theory suggests that imposing a new tax or raising an old one can backfire, resulting in insufficient or no gains in government revenues due to the decline in demand for the goods or services being taxed. A deadweight loss, therefore, disrupts the balance between supply and demand. English economist Alfred Marshall is widely credited as the originator of deadweight loss analysis.

Here's how it works. When the government raises taxes on certain goods and services, it collects that tax as additional revenue. Taxes, though, result in a higher cost of production and a higher purchase price for the consumer. This, in turn, causes production volumes (and, therefore, supply) to drop, leading to a drop in demand for these goods and services. This gap between the taxed and tax-free production volumes is the deadweight loss.

Open-Economy Macroeconomics: Basic Concepts

Open and Closed Economies

A *closed economy* is one that does not interact with other economies in the world. There are no exports, no imports, and no capital flows.

An *open economy* is one that interacts freely with other economies around the world. An open economy interacts with other countries in two ways.

1. It buys and sells goods and services in world product markets.
2. It buys and sells capital assets in world financial markets.

The Flow of Goods: Exports, Imports, Net Exports

Exports are goods and services that are produced domestically and sold abroad. *Imports* are goods and services that are produced abroad and sold domestically.

Net exports (NX) are the value of a nation's exports minus the value of its imports. Net exports are also called the *trade balance*.

A *trade deficit* is a situation in which net exports (*NX*) are negative. ($\text{Imports} > \text{Exports}$)

A *trade surplus* is a situation in which net exports (*NX*) are positive. ($\text{Exports} > \text{Imports}$)

Balanced trade refers to when net exports are zero—exports and imports are exactly equal.

- Factors That Affect Net Exports
- The tastes of consumers for domestic and foreign goods.

- The prices of goods at home and abroad.
- The exchange rates at which people can use domestic currency to buy foreign currencies.





The incomes of consumers at home and abroad.

- The costs of transporting goods from country to country.
- The policies of the government toward international trade.

The Prices for International Transactions: Real and Nominal Exchange Rates

International transactions are influenced by international prices. The two most important international prices are the nominal exchange rate and the real exchange rate.

Nominal Exchange Rates

The *nominal exchange rate* is the rate at which a person can trade the currency of one country for the currency of another. The nominal exchange rate is expressed in two ways:

In units of foreign currency per one U.S. dollar. And in units of U.S. dollars per one unit of the foreign currency. Assume the exchange rate between the Japanese yen and U.S. dollar is 80 yen to one dollar. One U.S. dollar trades for 80 yen. One yen trades for $1/80$ ($= 0.0125$) of a dollar.

- Appreciation* refers to an increase in the value of a currency as measured by the amount of foreign currency it can buy.
- Depreciation* refers to a decrease in the value of a currency as measured by the amount of foreign currency it can buy. If a dollar buys more foreign currency, there is an appreciation of the dollar. If it buys less there is a depreciation of the dollar.

Real Exchange Rates

The **real exchange rate** is the rate at which a person can trade the goods and services of one country for the goods and services of another.

The real exchange rate compares the prices of domestic goods and foreign goods in the domestic economy. If a case of German beer is twice as expensive as American beer, the real exchange rate is 1/2 case of German beer per case of American beer. The real exchange rate depends on the nominal exchange rate and the prices of goods in the two countries measured in local currencies. The real exchange rate is a key determinant of how much a country exports and imports.

$$\text{Real exchange rate} = \frac{\text{Nominal exchange rate} \times \text{Domestic price}}{\text{Foreign price}}$$

- A **depreciation (fall)** in the U.S. real exchange rate means that U.S. goods have become cheaper relative to foreign goods. This encourages consumers both at home and abroad to buy more U.S. goods and fewer goods from other countries. As a result, U.S. exports rise, and U.S. imports fall, and both of these changes raise U.S. net exports.
- Conversely, an **appreciation in the U.S.** real exchange rate means that U.S. goods have become more expensive compared to foreign goods, so U.S. net exports fall.

Fixed & floating Exchange Rates:



International currency exchange rates display how much one unit of a currency can be exchanged for another currency. Currency exchange rates can be floating, in which case they change continually based on a multitude of factors, or they can be pegged (or fixed) to another currency, in which case they still float, but they move in parallel with the currency to which they are pegged.

Differences of Fixed & Floating exchange Rate: A floating exchange rate is determined by the private market through supply and demand. A fixed, or pegged, rate is a rate the government (central bank) sets and maintains as the official exchange rate. The reasons to peg a currency are linked to stability. Especially in today's developing nations, a country may decide to peg its currency to create a stable atmosphere for foreign investment.

What Is the Balance of Payments (BOP)

The balance of payments (BOP), also known as the balance of international payments, is a statement of all transactions made between entities in one country and the rest of the world over a defined period, such as a quarter or a year. It summarizes all transactions that a country's individuals, companies, and government bodies complete with individuals, companies, and government bodies outside the country.

- The balance of payments includes both the current account and capital account.
- The current account includes a nation's net trade in goods and services, its net earnings on cross-border investments, and its net transfer payments.
- The capital account consists of a nation's transactions in financial instruments and central bank reserves.

What Are Terms of Trade (TOT)

Terms of trade (TOT) represent the ratio between a country's export prices and its import prices. How many units of exports are required to purchase a single unit of imports? The ratio is calculated by dividing the price of the exports by the price of the imports and multiplying the result by 100.

What Is the Balance of Trade (BOT)

Balance of trade (BOT) is the difference between the value of a country's exports and the value of a country's imports for a given period. Balance of trade is the largest component of a country's balance of payments (BOP). Sometimes the balance of trade between a country's goods and the balance of trade between its services are distinguished as two separate figures.

- A country that imports more goods and services than it exports in terms of value has a trade deficit while a country that exports more goods and services than it imports has a trade surplus.
- The balance of trade is also referred to as the trade balance, the international trade balance, commercial balance, or the net exports.

Trade Surplus & Trade Deficit:

A trade surplus is an economic measure of a positive balance of trade, where a country's exports exceed its imports.

$$\square \quad \text{Trade Balance} = \text{Total Value of Exports} - \text{Total Value of Imports}$$

Trade Deficit:

A trade deficit occurs when a country's imports exceed its exports during a given time period. It is also referred to as a negative balance of trade (BOT).

The balance can be calculated on different categories of transactions: goods (a.k.a., “merchandise”), services, goods and services. Balances are also calculated for international transactions—current account, capital account, and financial account.

Inflation

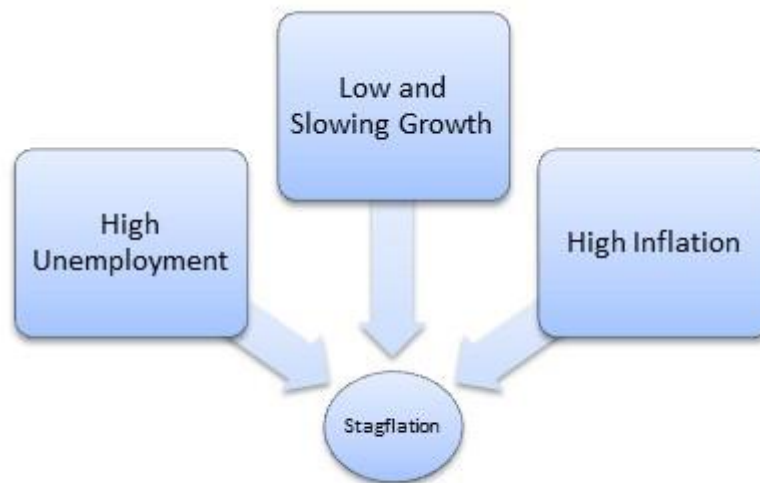
Inflation is a rise in the general level of prices of goods and services in an economy over a period of time.

When the general price level rises, each unit of currency buys fewer goods and services. Therefore, inflation also reflects a loss of purchasing power of money.

In Economics, the word ‘inflation’ refers to General rise in Prices Measured against a Standard Level of Purchasing Power.

Here are several variations on inflation used popularly to indicate specific meanings.

- **Deflation** is when the general level of prices is falling. It is the opposite of inflation. Also referred to as Deflation. The lack of inflation may be an indication that the economy is weakening.
- **Hyperinflation** is unusually rapid inflation in very short span of time. In extreme cases, this can lead to the breakdown of a nation’s monetary system with complete loss of confidence in the domestic currency. One of the earlier examples of hyperinflation occurred in Germany in early 1920 s after the First World War, when prices rose 2,500% in one month.
- **Stagflation** is the combination of high unemployment with high inflation. This happened in industrialized countries during the 1970s, when a bad economy was combined with OPEC raising oil prices led to low growth.



Inflation is all about prices going up, but for healthy economy wages should be rising as well. The question shouldn't be whether inflation is rising, but whether it's rising at a quicker pace than your wages, if the answer is a Yes only then inflation is problematic.

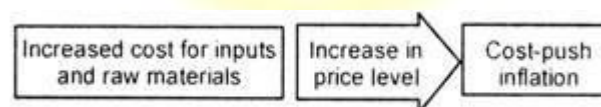
3 types of inflation:

The three types of inflation are demand-pull, cost-push and built-in inflation.

- a. **Demand-pull inflation:** it occurs when the demand for goods or services is higher when compared to the production capacity. The difference between demand and supply (shortage) result in price appreciation.



- b. **Cost-push inflation:** it occurs when the cost of production increases. Increase in prices of the inputs (labour, raw materials, etc.) Increases the price of the product. Cost of production may rise due to an increase in the prices of raw materials, wages, etc. Higher wage means high cost of production. Prices of commodities are thereby increased.



- c. **Built-in inflation:** expectation of future inflations results in built-in inflation. A rise in prices results in higher wages to afford the increased cost of living. Therefore, high wages result in increased cost of production, which in turn has an impact on product pricing. The circle hence continues.

What are the main causes of inflation?

1. **Monetary policy:** it determines the supply of currency in the market. Excess supply of money leads to inflation. Hence decreasing the value of the currency.



2. **Fiscal policy:** it monitors the borrowing and spending of the economy. Higher borrowings (debt), result in increased taxes and additional currency printing to repay the debt.
3. **Demand-pull inflation:** increases in prices due to the gap between the demand (higher) and supply (lower).
4. **Cost-push inflation:** higher prices of goods and services due to increased cost of production.
5. **Exchange rates:** exposure to foreign markets are based on the dollar value. Fluctuations in the exchange rate have an impact on the rate of inflation.

Who benefits from inflation?

Inflation being a cause of concern for the economy, doesn't affect everyone in a bad way. It is a boon for a certain set of people. While consumers lose a part of their purchasing power to inflation, investors gain from it.

Investors investing in assets affected by inflation, if held on for a long time will certainly benefit from it. For example, an increase in housing prices might affect consumers. However, those who have already bought a house will benefit from capital appreciation.

How do we prevent inflation?

Inflation is generally controlled by the Central Bank and/or the government. The main policy used is monetary policy (changing interest rates). However, in theory, there are a variety of tools to control inflation including:

1. **Monetary policy** – Higher interest rates reduce demand in the economy, leading to lower economic growth and lower inflation. A higher interest rate should also lead to a higher exchange rate, which helps to reduce inflationary pressure by:
 - Making imports cheaper. (lower price of imported goods) □
Reducing demand for exports.
 - Increasing incentive for exporters to cut costs.
2. **Control of money supply** – Monetarists argue there is a close link between the money supply and inflation, therefore controlling money supply can control inflation.
Monetarists believe there is a strong link between the money supply and inflation. If you can control the growth of the money supply, then you should be able to bring inflation under control. Monetarists would stress policies such as:
 - Higher interest rates (tightening monetary policy)
 - Reducing budget deficit (deflationary fiscal policy)
 - Control of money being created by the government

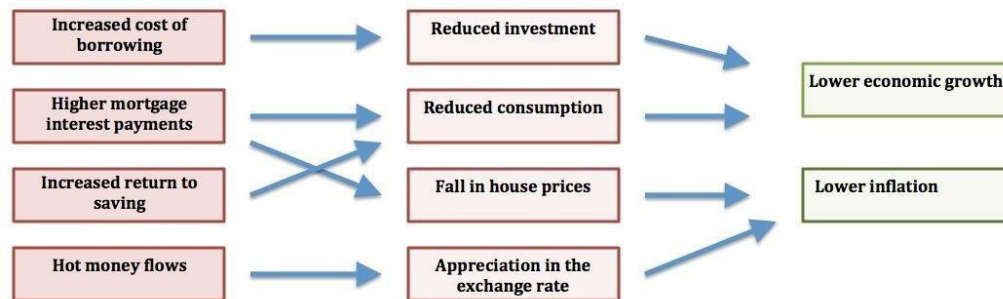
3. **Supply-side policies** – policies to increase the competitiveness and efficiency of the economy, putting downward pressure on long-term costs.
4. **Fiscal policy** – a higher rate of income tax could reduce spending, demand and inflationary pressures.

The government can increase taxes (such as income tax and VAT) and cut spending. This improves the government's budget situation and helps to reduce demand in the economy. Both these policies reduce inflation by reducing the growth of aggregate demand.



5. **Wage controls** – trying to control wages could, in theory, help to reduce inflationary pressures. However, apart from the 1970s, it has been rarely used.

Effect of higher interest rates



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The Consumer Price Index

The Consumer Price Index (CPI) is a measure that examines the weighted average of prices of a basket of consumer goods and services, such as transportation, food, and medical care. It is calculated by taking price changes for each item in the predetermined basket of goods and averaging them. Changes in the CPI are used to assess price changes associated with the cost of living.

The CPI is one of the most frequently used measures of inflation and deflation. It may be compared with the producer price index (PPI), which instead of considering prices paid by consumers looks at what businesses pay for inputs.

Calculating CPI

The formula used to calculate the Consumer Price Index for a single item is as follows:

$$\text{CPI} = \frac{\text{Cost of the Market Basket in Given Year}}{\text{Cost of the Market Basket in Base Year}} \times 100\%$$

The CPI tells you about the *value* of the money in your pocket. In Table 22.1, you established that in 2010 (the base period), the cost of the CPI basket was \$ 50 and in 2011, it was \$70. If we use these numbers in the CPI formula, we can find the CPI for 2010 and 2011. For 2010, the CPI is

$$\text{CPI in 2010} = \frac{\$50}{\$50} \times 100 = 100.$$

For 2011, the CPI is

$$\text{CPI in 2011} = \frac{\$70}{\$50} \times 100 = 140.$$

TABLE 22.1 The CPI:
A Simplified Calculation

(a) The cost of the CPI basket at base-period prices: 2010

CPI basket		Price	Cost of CPI Basket
Item	Quantity		
Oranges	10	\$1.00	\$10
Haircuts	5	\$8.00	<u>\$40</u>
Cost of CPI basket at base-period prices			<u><u>\$50</u></u>

(b) The cost of the CPI basket at current-period prices: 2011

CPI basket		Price	Cost of CPI Basket
Item	Quantity		
Oranges	10	\$2.00	\$20
Haircuts	5	\$10.00	<u>\$50</u>
Cost of CPI basket at current-period prices			<u><u>\$70</u></u>

Measuring the Inflation Rate

Inflation is measured by calculating the percentage rate of change of a price index, which is called the inflation rate.

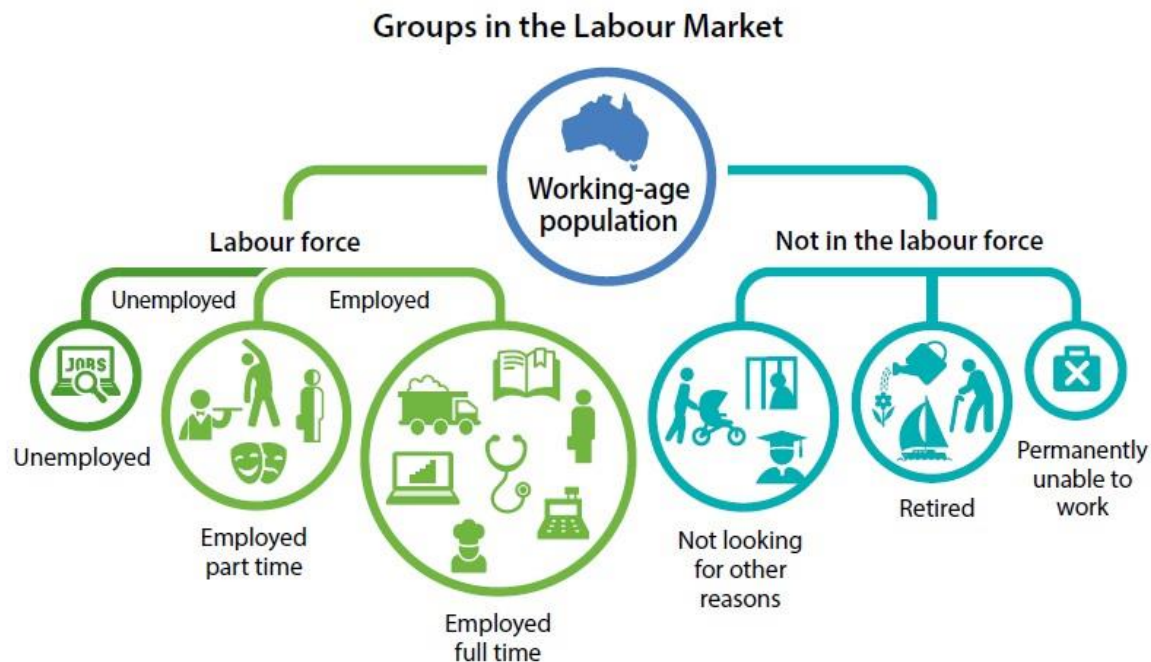
A major purpose of the CPI is to measure changes in the cost of living and in the value of money. To measure these changes, we calculate the *inflation rate* as the annual percentage change in the CPI. To calculate the inflation rate, we use the formula:

$$\text{Inflation rate} = \frac{\text{CPI this year} - \text{CPI last year}}{\text{CPI last year}} \times 100.$$

We can use this formula to calculate the inflation rate in 2010. The CPI in June 2010 was 218.0, and the CPI in June 2009 was 215.7. So the inflation rate during the twelve months to June 2010 was

$$\text{Inflation rate} = \frac{(218.0 - 215.7)}{215.7} \times 100 = 1.1\%.$$

Unemployment:



As part of this survey, people aged 15 years and over (the working-age population) into three broad categories:

- ☐ Employed _ includes people who are in a paid job for one hour or more in a week.
- ☐ Unemployed _ includes people who are not in a paid job, but who are actively looking for work.
- ☐ Not in the labour force _ includes people not in a paid job, and who are not looking for work.

Once the number of people in each of these categories has been estimated, the following labour market indicators can be calculated:

- ☐ Labour force _ the sum of employed and unemployed people.

$$\text{Labor-force participation rate} = 100 \times \frac{\text{Labor force}}{\text{Adult population}}$$

- ☐ Unemployment rate _ the percentage of people in the labour force that are unemployed.

$$\text{Unemployment rate} = \frac{\text{number of unemployed people}}{\text{labor force}}$$

- ☐ Participation rate _ the percentage of people in the working-age population that are in the labour force.

$$\boxed{\text{Participation Rate}} = \frac{\boxed{\text{Labour force}}}{\boxed{\text{Working-age population}}} \times 100$$



What are the main types of unemployment?

There are three main types of unemployment – cyclical, structural and frictional unemployment. In practice, these cannot be measured directly, and they can often overlap, but they provide a useful way of thinking about unemployment.

1. Cyclical Unemployment

Cyclical unemployment occurs with changes in economic activity over the business cycle. During an economic downturn, a shortfall of demand for goods and services results in a lack of jobs being available for those who want to work. Businesses experiencing weaker demand might reduce the amount of people they employ by laying off existing workers, or hiring fewer new workers. As a result, people looking for work will also find it harder to become employed.

An increase in cyclical unemployment might suggest the economy is operating below its potential. With more people competing for jobs, businesses might offer lower wage increases, which would contribute to lower inflation.

Policies that stimulate aggregate demand, such as expansionary monetary policy, can help reduce this type of unemployment (because businesses experiencing stronger demand are likely to employ more people).

2. Structural Unemployment

Structural unemployment occurs when there is a mismatch between the jobs that are available and the people looking for work. This mismatch could be because jobseekers don't have the skills required to do the available jobs, or because the available jobs are a long way from the jobseekers. Workers may become unemployed if they work in industries that are declining in size or have skills that could be automated as a result of large-scale technological advances. It may be difficult for them to find work in another industry and they may need to develop new skills or move to a region that has more opportunities. Structural unemployment tends to be longer lasting than other types of unemployment.

This is because it can take a number of years for workers to develop new skills or move to a different region to find a job that matches their skills. As a result, workers who are unemployed because of structural factors are more likely to face long-term unemployment (for more than 12 months).

In theory, this type of unemployment should not directly influence wages or inflation and is best addressed through policies that focus on skills and the supply of labour.

3. Frictional Unemployment

Frictional unemployment occurs when people move between jobs in the labour market, as well as when people transition into and out of the labour force.

However, people may not find jobs immediately and need to invest time and effort in searching for the right job. Businesses also spend time searching for suitable candidates to fill job vacancies.

As a result, people looking for jobs are not matched immediately with vacancies and may experience a period of temporary unemployment.



This type of unemployment is generally shorter term (less than one month). Frictional unemployment is likely to occur at all points of the business cycle and, like structural unemployment, may not influence wages or inflation.

Other Types of Unemployment

Underemployment occurs when people are employed, but would like and are available to work more hours. There are two categories,

- First, part-time workers who would prefer to work additional hours.
- Second, people who usually work full time, but are currently working part-time hours.

Underemployment rates are generally higher among groups that have a larger proportion of people working part time, such as females, younger workers and older workers.

Hidden unemployment occurs when people are not counted as unemployed in the formal labour market statistics, but would probably work if they had the chance. For example, someone might have looked for work for a long time, given up hope and stopped looking, but still wish to work. (These people are sometimes referred to as 'discouraged workers'.)

Seasonal unemployment occurs at different points over the year because of seasonal patterns that affect jobs. Some examples include ski instructors, fruit pickers and holiday-related jobs.

Business Cycle:

A business cycle is a cycle of fluctuations in the Gross Domestic Product (GDP) around its long-term natural growth rate. It explains the expansion and contraction in economic activity that an economy experiences over time.

Prof. James Astray:

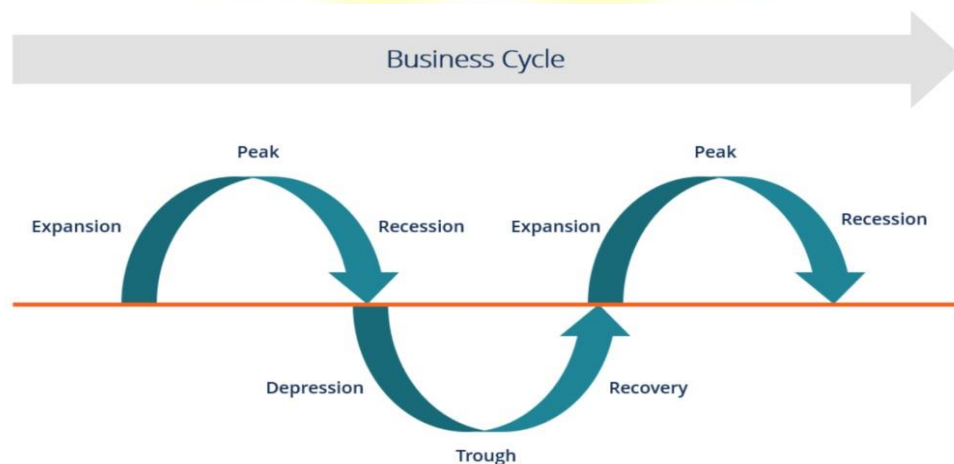
Fluctuations in all economic activities is called trade cycle.

Prof. Habler:

Trade cycle is a name of good and bad trade phase.

Prof. W.C Mitchel:

Fluctuations in aggregate economics activities of nation is called trade cycle.





A business cycle is completed when it goes through a single boom and a single contraction in sequence. The time period to complete this sequence is called the length of the business cycle. A boom is characterized by a period of rapid economic growth whereas a period of relatively stagnated economic growth is a recession. These are measured in terms of the growth of the real GDP, which is inflation-adjusted.

Stages of the Business Cycle

In the diagram above, the straight line in the middle is the steady growth line. The business cycle moves about the line. Below is a more detailed description of each stage in the business cycle:

1. Expansion

The first stage in the business cycle is expansion. In this stage, there is an increase in positive economic indicators such as employment, income, output, wages, profits, demand, and supply of goods and services. Debtors are generally paying their debts on time, the velocity of the money supply is high, and investment is high. This process continues as long as economic conditions are favorable for expansion.

2. Peak

The economy then reaches a saturation point, or peak, which is the second stage of the business cycle. The maximum limit of growth is attained. The economic indicators do not grow further and are at their highest. Prices are at their peak. This stage marks the reversal point in the trend of economic growth. Consumers tend to restructure their budgets at this point.

3. Recession

The recession is the stage that follows the peak phase. The demand for goods and services starts declining rapidly and steadily in this phase. Producers do not notice the decrease in demand instantly and go on producing, which creates a situation of excess supply in the market. Prices tend to fall. All positive economic indicators such as income, output, wages, etc., consequently start to fall.

4. Depression

There is a commensurate rise in unemployment. The growth in the economy continues to decline, and as this falls below the steady growth line, the stage is called a depression.

5. Trough

In the depression stage, the economy's growth rate becomes negative. There is further decline until the prices of factors, as well as the demand and supply of goods and services, contract to reach their lowest point. The economy eventually reaches the trough. It is the negative saturation point for an economy. There is extensive depletion of national income and expenditure.

6. Recovery

After the trough, the economy moves to the stage of recovery. In this phase, there is a turnaround in the economy, and it begins to recover from the negative growth rate. Demand starts to pick up

due to low prices and, consequently, supply begins to increase. The population develops a positive attitude towards investment and employment and production starts increasing.





A central bank is an independent national authority that conducts monetary policy, regulates banks, and provides financial services, including economic research. Its goals are to stabilize the nation's currency, keep unemployment low, and prevent inflation.

Definition and Example of a Central Bank

Though they may be established by a governing body, central banks are independent authorities. They have a number of duties related to monetary policy, providing financial services, regulating lower banks, and conducting research. Central banks aim to keep a nation's currency and economy stable.

State Bank of Pakistan, Reserve Bank of India, The Bank of England and U.S. Federal Reserve is another example.

How Central Banks Work**1. Bank Regulation**

Central banks regulate their members. They require enough reserves to cover potential loan losses. They are responsible for ensuring financial stability and protecting depositors' funds.

2. Providing Financial Services

Central banks serve as the bank for private banks and the nation's government. They process checks and lend money to their members.

3. Central banks store currency in their foreign exchange reserves. They use these reserves to change exchange rates. They add foreign currency, usually the dollar or euro, to keep their own currency in alignment. That's called a peg, and it helps exporters keep their prices competitive.**4. Central banks also regulate exchange rates as a way to control inflation. They buy and sell large quantities of foreign currency to affect supply and demand.****5. Most central banks produce regular economic statistics to guide fiscal policy decisions. Here are examples of reports provided by the Federal Reserve.****6. Monetary Policy**

Central banks affect economic growth by controlling the liquidity in the financial system. They have three monetary policy tools to achieve this goal.

- Reserve Requirement
- Quantitative Easing
- Interest Rates They Charge Their Member Banks

What is Monetary Policy?

Monetary policy is an economic policy that manages the size and growth rate of the money supply in an economy. It is a powerful tool to regulate macroeconomic variables such as inflation and unemployment.



These policies are implemented through different tools, including the adjustment of the interest rates, purchase or sale of government securities, and changing the amount of cash circulating in the economy. The central bank or a similar regulatory organization is responsible for formulating these policies.

Objectives of Monetary Policy

The primary objectives of monetary policies are the management of inflation or unemployment, and maintenance of currency exchange rates.

1. Inflation

Monetary policies can target inflation levels. A low level of inflation is considered to be healthy for the economy. If inflation is high, a contractionary policy can address this issue.

2. Unemployment

Monetary policies can influence the level of unemployment in the economy. For example, an expansionary monetary policy generally decreases unemployment because the higher money supply stimulates business activities that lead to the expansion of the job market.

- Keeping in view the present situation of unemployment and disguised unemployment particularly in more growing populated countries, the said objective of monetary policy is most suitable.
- On humanitarian grounds, the policy can go a long way to solve the acute problem of unemployment.
- It is useful tool to provide economic and social welfare of the community.

3. Currency exchange rates

Using its fiscal authority, a central bank can regulate the exchange rates between domestic and foreign currencies. For example, the central bank may increase the money supply by issuing more currency. In such a case, the domestic currency becomes cheaper relative to its foreign counterparts.

- It leads to violent fluctuations resulting in encouragement to speculative activities in the market.
- Heavy fluctuations lead to loss of confidence on the part of domestic and foreign capitalists resulting in adverse impact in capital outflow which may also result in capital formation and growth.
- Fluctuations in exchange rates bring repercussions in the internal price level.

4. Economic Growth:

Therefore, monetary policy promotes sustained and continuous economic growth by maintaining equilibrium between the total demand for money and total production capacity and further creating favourable conditions for saving and investment. For bringing equality between demand and supply, flexible monetary policy is the best course.

5. Neutrality of Money:

Economists like Wicksteed, Hayek and Robertson are the chief exponents of neutral money. They hold the view that monetary authority should aim at neutrality of money in the economy. Any monetary change is the root cause of all economic fluctuations.



They are of the confirmed view that if somehow neutral monetary policy is followed, there will be no cyclical fluctuations, no trade cycle, no inflation and no deflation in the economy. Under this system, money is kept stable by the monetary authority.

6. Price Stability:

The objective of price stability has been highlighted during the twenties and thirties of the present century. In fact, economists like Crustar Cassels and Keynes suggested price stabilization as a main objective of monetary policy. It promotes business activity and ensures equitable distribution of income and wealth. As a consequence, there is general wave of prosperity and welfare in the community.

Tools of Monetary Policy

Central banks have four main monetary policy tools: the reserve requirement, open market operations, and interest on reserves

Most central banks also have a lot more tools at their disposal. Here are the four primary tools and how they work together to sustain healthy economic growth.

1. Interest rate adjustment

A central bank can influence interest rates by changing the discount rate. The discount rate (base rate) is an interest rate charged by a central bank to banks for short-term loans. For example, if a central bank increases the discount rate, the cost of borrowing for the banks increases. Subsequently, the banks will increase the interest rate they charge their customers. Thus, the cost of borrowing in the economy will increase, and the money supply will decrease.

2. Change reserve requirements

Central banks usually set up the minimum amount of reserves that must be held by a commercial bank. By changing the required amount, the central bank can influence the money supply in the economy. If monetary authorities increase the required reserve amount, commercial banks find less money available to lend to their clients and thus, money supply decreases. Commercial banks can't use the reserves to make loans or fund investments into new businesses. Since it constitutes a lost opportunity for the commercial banks, central banks pay them interest on the reserves. The interest is known as IOR or IORR (interest on reserves or interest on required reserves).

3. Open market operations

Open market operations are when central banks buy or sell securities. These are bought from or sold to the country's private banks. When the central bank buys securities, it adds cash to the banks' reserves. That gives them more money to lend. When the central bank sells the securities, it places them on the banks' balance sheets and reduces its cash holdings. The bank now has less to lend. A central bank buys securities when it wants an expansionary monetary policy. It sells them when it executes tight monetary policy.

The central bank can either purchase or sell securities issued by the government to affect the money supply. For example, central banks can purchase government bonds. As a result, banks will obtain more money to increase the lending and money supply in the economy.

How These Tools Work

Central bank tools work by increasing or decreasing total liquidity. That's the amount of capital available to invest or lend. It's also money and credit that consumers spend. It's technically more than the money supply, known as M1 and M2. The M1 symbol denotes currency and check deposits. M2 is money market funds, CDs, and savings accounts. Therefore, when people say that central bank tools affect the money supply, they are understating the impact.

Expansionary vs. Contractionary Monetary Policy

Depending on its objectives, monetary policies can be expansionary or contractionary.

Expansionary Monetary Policy:

This is a monetary policy that aims to increase the money supply in the economy by decreasing interest rates, purchasing government securities by central banks, and lowering the reserve requirements for banks. An expansionary policy lowers unemployment and stimulates business activities and consumer spending. The overall goal of the expansionary monetary policy is to fuel economic growth. However, it can also possibly lead to higher inflation.

Contractionary Monetary Policy:

The goal of a contractionary monetary policy is to decrease the money supply in the economy. It can be achieved by raising interest rates, selling government bonds, and increasing the reserve requirements for banks. The contractionary policy is utilized when the government wants to control inflation levels.

Fiscal Policy:

Fiscal policy is the means by which a government adjusts its spending levels and tax rates to monitor and influence a nation's economy. It is the sister strategy to monetary policy through which a central bank influences a nation's money supply.

- Fiscal policy is the means by which a government adjusts its spending levels and tax rates to monitor and influence a nation's economy.
- It is the sister strategy to monetary policy through which a central bank influences a nation's money supply.
- Using a mix of monetary and fiscal policies, governments can control economic phenomena.

How Fiscal Policy Works

Fiscal policy is based on the theories of British economist John Maynard Keynes. Also known as Keynesian economics, this theory basically states that governments can influence macroeconomic productivity levels by increasing or decreasing tax levels and public spending.

Balancing Act

The idea is to find a balance between tax rates and public spending. For example, stimulating a stagnant economy by increasing spending or lowering taxes, also known as expansionary fiscal

policy, runs the risk of causing inflation to rise. This is because an increase in the amount of money in the economy, followed by an increase in consumer demand, can result in a decrease in



the value of money—meaning that it would take more money to buy something that has not changed in value.

Let's say that an economy has slowed down. Unemployment levels are up, consumer spending is down, and businesses are not making substantial profits. A government may decide to fuel the economy's engine by decreasing taxation, which gives consumers more spending money while increasing government spending in the form of buying services from the market (such as building roads or schools). By paying for such services, the government creates jobs and wages that are in turn pumped into the economy. Pumping money into the economy by decreasing taxation and increasing government spending is also known as "pump priming." In the meantime, overall unemployment levels will fall.

Kinds of Fiscal Policy:

There are two kinds of fiscal policy which is contractionary and expansionary fiscal policy.

Contractionary fiscal policy is said to be in action when the government reduces spending and increases the taxes at the same time in the country.

- The result of such a move is that there is very less money available in the market. It leads to reduction in the purchasing power which results in declining consumption.
- As less capital is available for business, the economy contracts and also causes unemployment. Contractionary policy is used to control inflation.

Expansionary fiscal policy is said to be in action when the government increases the spending and lowers tax rates for boosting economic growth.

- This increases consumption as there is a rise in purchasing power.
- Businesses get easy access to credit and therefore invest in new projects and thus, GDP of the nation is increased.

Fiscal Policy Tools

When deciding fiscal policy, government officials have two tools that they can use. They are taxation and spending. Whilst they are somewhat interlinked, they are not completely dependent on each other.

For example, governments frequently spend more than they bring in through taxation. Also known as a budget deficit, the fiscal policy does not consider the need to raise taxes to fund expenditure.

1. Taxation as a Fiscal Tool

Governments use taxation as a way of funding expenditures. However, higher taxes are generally unpopular with voters. Yet they generally want greater spending in areas such as education, defence, and healthcare. So there is a tough balancing act that most governments don't follow; frequently spending more than they receive in taxes.

The 'big' types of taxation such as sales and income are generally left untouched. However, governments have developed other means that are less politically sensitive. For instance, inheritance tax, corporation tax, land tax, and so on. Consequently, these methods are used to essentially control their expenditure, allowing greater flexibility at a lower political cost. However, higher taxes do not necessarily mean higher revenue.





There are also several other non-revenue objectives of taxation:

1. Regulating Consumption

Governments may look to dis-incentivize the consumption of demerit goods such as cigarettes or sugary soft drinks.

2. Protecting Domestic Industries

Tariffs or taxes on imported goods make international alternatives more expensive. In turn, domestic companies can compete more effectively on price.

3. Encourage Investment

Surcharges on goods such as oil and gas make it more expensive for consumers. In turn, this can impact on consumption, thereby reducing profits. Such companies may then look to diversify into other areas such as renewable energy where the tax regime is not so strict.

4. Reduce Inequality

Perhaps one of the most prominent aims in society today is for taxation to reduce inequality.

2. Spending as a Fiscal Tool

Government spending is another type of fiscal policy and plays an important part in shaping the overall economy.

Aims of Government Spending

The general aim of government spending is to deliver on its key responsibilities such as defence, and law and order. However, over the past 100 years, it has developed into a social structure to protect the poor, vulnerable, and needy.

The aims of government spending developed from the desire to obtain social justice and redistribution. They can be summarized below:

1. Reduce Inequality

Social security is a prominent method by which governments aim to reduce inequality through redistribution.

2. Public Support

Governments spend money in order to gain public support. If the public see more 'free' public services opening, they are more likely to favour the existing government.

3. Increase Economic Growth

Higher government spending has the potential to increase economic growth. However, it depends at what point of the business cycle it occurs at. For instance, when going through an economic boom, it is unlikely to help as it starves private firms of capital. At the same time, if spending is stimulated through borrowing, it may improve the economy in the short-term, but will cause greater problems in the long term.

4. Defence

There is a constant global threat from international powers. Not only physically, but also in terms of intellectual property and other non-physical attacks. For instance, cyber-crime has become an increasingly prominent issue in national defence.

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5. Law and Order

Traditionally seen as a service that the private sector would be unable to offer. Therefore, it is in the interest of all parties to ensure there is a deterrent.

Objectives of Fiscal Policy

Governments have several objectives in mind when deciding on fiscal policy. The six main ones are listed below. However, they are not necessarily considered in the same fashion. For instance, some governments, like Greece, may be forced into controlling debt over full employment. In the same fashion, a country in a sound financial situation may wish to target full employment over controlling its small level of debt.

Governments may favour one objective over the others. So let us look at what the six main objectives of fiscal policy are.

1. Full Employment

One of the governments' main objectives is to keep and get people into work. Not only do governments benefit from higher taxes, but also from lower expenditures on social security. An expansionary policy may look to invest in infrastructure which would directly create employment. Alternatively, it may reduce taxes to give consumers more money to indirectly stimulate employment from their purchases.

2. Economic Growth

As an economy grows, its citizens, on the whole, become more prosperous. So this is an important objective. At the same time, governments must be careful. An aggressive expansionary fiscal policy could prove detrimental in the long-term. We only need to look at Greece as an example. Whilst economic growth is an understandable aim, it must also be considered alongside stability. For instance, heavy government spending can contribute to inflation. Too much spending and it can create high levels of debt and an inflation rate that destroys the nation's wealth.

3. Control Debt

Running a budget deficit is not necessarily bad. However, over time it creates more and more debt. If economic growth and tax receipts do not increase it line, a nation faces an unsustainable level of debt. A rational fiscal policy would aim to control this before having to take drastic action.

4. Control Inflation

When an economy is growing strongly, it may experience high levels of inflation (this may also depend on monetary policy). As 2 percent is generally the target, anything above this is a cause for concern: particularly if it is consistently above. Even though inflation is a monetary phenomenon, there are steps governments take to try and stem such. With that said, there is little fiscal policy can do if the money supply has been let loose. Nevertheless, governments try by increasing taxes to reduce disposable incomes and hence consumption.

5. Re-distribution

Another aim of the government is to transfer wealth from the rich to the poor. Higher taxes on the rich can sometimes result in high tax receipts, but this is not always the case. Evasion and

avoidance may occur, or they may in fact just leave the country. Although small incremental increases may not have such a significant impact in the short-term.

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6. Political

Fiscal policy has inevitably become a political tool that many incumbent governments use in a bid to get re-elected. A loosening of fiscal policy and greater spending can often win some 'floating voters' over.

Difference b/w Fiscal Deficit & Fiscal surplus:

- **Fiscal Deficit:** If govt. spends more than what it earns, it leads to the deficit.
- **Fiscal surplus:** If the govt. spends less than what it earns, it creates a fiscal surplus.

Free Trade & Protection (WTO):

Free trade, also called **laissez-faire**, a policy by which a government does not discriminate against imports or interfere with exports by applying tariffs (to imports) or subsidies (to exports). A free-trade policy does not necessarily imply, however, that a country abandons all control and taxation of imports and exports.

A free trade agreement is a pact between two or more nations to reduce barriers to imports and exports among them. Under a free trade policy, goods and services can be bought and sold across international borders with little or no government tariffs, quotas, subsidies, or prohibitions to inhibit their exchange.

The WTO is sometimes described as a “free trade” institution, but that is not entirely accurate. The system does allow tariffs and, in limited circumstances, other forms of protection. More accurately, it is a system of rules dedicated to open, fair and undistorted competition.

The concept of free trade is the opposite of trade protectionism or economic isolationism.

Protectionism, policy of protecting domestic industries against foreign competition by means of tariffs, subsidies, import quotas, or other restrictions or handicaps placed on the imports of foreign competitors. Protectionist policies have been implemented by many countries despite the fact that virtually all mainstream economists agree that the world economy generally benefits from free trade.